MILITARY SPECIFICATION

HANDLE, OXYACETYLENE WELDING AND CUTTING TORCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers replacement handles for hand held oxyacetylene torch used for welding and cutting ferrous metals.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document ImprovementProposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3433
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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COMPRESSED GAS ASSOCIATION (CGA)

CGA E-1 - Standard Connections for Regulator Outlets, Torches and Fitted Hose for Welding and Cutting Equipment
CGA E-2 - Hose Line Check Valve Standards for Welding and Cutting
CGA E-5 - Torch Standard for Welding and Cutting

(Requests for copies should be addressed to the Compressed Gas Association, Inc., 1235 Jefferson Davis Hwy, Arlington, VA 22202.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2.1 and 6.3).

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3.2 Design. The torch handle shall be a replacement for the brand name and model number of handle as specified in the contract (see 6.2.1) and shall meet the requirements of CGA E-5 for hand held oxyacetylene torches of the medium (equal volume) pressure class. Medium pressure is acetylene pressure from one to fifteen pounds per square inch. The torch handle shall be designed to fit and perform with the mating torch parts as specified in the contract or purchase order (see 6.2.1) and in accordance with the requirements of this specification.

3.2.1 Construction. All parts used in fabrication of the handles shall be new, clean, and free of oil and grease. In no event shall patching processes be used for reclaiming defective parts. Surfaces of all parts shall be free of burrs, defects, and dirt. Parts subject to removal or adjustment shall not be permanently deformed by the assembly process.

3.2.2 Threads. Screw threads shall comply with the requirements of FED-STD-H28 and the Detailed Standard sections referenced therein. All internal and external threads shall be clean, free of any extraneous materials, and shall have no burrs or nicks.

3.2.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the handle. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.4 Brazing. Brazing shall be of the highest quality workmanship prevailing among commercial manufacturers normally producing a handle of the types specified herein. Flux shall be cleaned from surfaces after brazing is completed. Brazing filler metal shall have a melting temperature no lower than 1000 degrees Fahrenheit (F). Filler metal shall be visibly complete around all brazed joints.

3.2.5 Machined surfaces. Gas holes and passages shall be free of burrs. Machined surface seals shall be free of nicks, burrs, and scratches and shall meet the leakage requirements of 3.3.3.

3.2.6 Lubricants. Lubricants inside the handle shall be compatible with oxygen at 90 pounds per square inch gauge (psig).

3.2.7 Materials. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3 Performance and product characteristics.

3.3.1 Oxygen flow rate. The handle shall have a cutting oxygen flow capacity of 450 standard cubic feet per hour (scfh) or more at an oxygen inlet pressure not exceeding 90 psig.

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3.3.2 Acetylene flow rate. The handle shall have an acetylene flow capacity of at least 25 scfh at an acetylene inlet pressure not exceeding 10 psig.

3.3.3 Leakage. When properly attached to the specified welding head mixer with a plugged tip and gas hoses, joints, valves, metal to metal mating seals, gas passages, connections, and o-ring glands in the handle shall sustain gas pressures of at least 50 psig internally (from one gas passage to another) and externally (from internal gas passage to external atmosphere). Valves shall sustain the gas pressures without leakage that is sufficient for formation of a bubble when submerged in clean water for ten seconds, both when the valves are closed and when the valves are open to gas passage.

3.4 Details of components. The handle shall consist of at least the components in 3.4.1 through 3.4.3.

3.4.1 Grip. The grip of the torch handle shall be knurled, grooved, or ribbed for at least four inches of its length. Grips that are part of a handle not constructed with gas passages drilled or formed through a solid body handle shall be fixed to the handle with pins, screws, threads, solder, or brazing metal.

3.4.2 Gas control valves. The oxygen and acetylene/fuel control valves shall be permanently and legibly marked on the valve body, the valve control knob, or the torch body or handle at a point adjacent to the inlet in accordance with the instruction for marking gas inlets specified in 3.4.2.5. If the valve knobs are marked, the valve stems shall not be interchangeable. Valve stems shall be equipped with packing and shall meet the leakage requirements of 3.3.3. The valve stem packing shall be on the down-stream side of the valve. Each control valve seat shall provide a seal that does not allow leak-through at 50 psig or more, from the up-stream side of the valve to the down-stream side of the valve, when the valve is closed with no more than 20 inch-pounds of torque applied to the knob. Valve seats shall be constructed so that seating occurs between materials of different hardness.

3.4.2.1 Valve bodies. Valve bodies shall be formed separately or as integral parts of the torch handle. Valve bodies which are formed separately shall be attached to the torch handle and shall meet the leakage requirements of 3.3.3.

3.4.2.2 Valve stems. Valve stems shall be fabricated of corrosion-resistant metal. Stems shall be assembled to prevent removal from the valve without first removing the gland nut and packing.

3.4.2.3 Valve packing. The valve packing shall not leak during torch operations. The packing, or alternate means, shall provide a drag equivalent of at least four inch-ounces of torque on the valve stem to aid the stem in maintaining its setting during use, and to deter the stem from being turned by accident.

3.4.2.4 Valve knobs. Valve knobs shall be fabricated of corrosion-resistant metal. Valve knobs shall be at least 0.875 inch in diameter and shall provide a positive grip for an operator wearing welding gloves. Valve knobs shall not separate from the stem when pulled with a force of at least 70 pounds along the centerline of the stem. Valve knobs shall not loosen or separate from the stem when at least 50 inch-pounds of torque is applied to the knob in both the clockwise and counter-clockwise directions.

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3.4.2.5 Gas inlets. Acetylene and oxygen inlets shall have threaded hose connections that comply with the requirements of CGA E-1, connections 022 and 023. Each inlet shall be permanently marked to indicate acetylene or fuel and oxygen. The minimum allowed marking shall be no less than an "A" or "F", and "O". Hose connections shall be at the rear of the torch handle. All threads shall be clean and free of dirt. Conical seats shall be smooth, symmetrical, and free of marks, nicks, and corrosion.

3.4.3 Reverse flow check valves. Two reverse flow check valves, conforming to CGA E-2, one for each inlet valve, shall be provided with each handle. When attached to the handle, the mating fit of the reverse flow check valves shall meet the requirements of 3.3.3.

3.5 Product identification. The handle shall be marked for product identification in accordance with MIL-STD-130. Unless otherwise specified (see 6.2.1), handle shall be permanently marked with the manufacturer's name and model number.

3.6 Workmanship. Workmanship shall be of the highest quality prevailing among manufacturers normally producing handles of the type specified herein for the commercial market.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When specified (see 6.2.1), first article inspection shall be performed on the preproduction model(s) or initial production item(s) noted in 3.1. First article inspection shall consist of the examination in 4.5, and the tests of 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

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4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production units offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (c) as follows, and failure of any unit to pass an examination or test shall be cause for rejection of the unit.

a. Product examination (see 4.5)

b. Leakage test (see 4.6.3)

c. Packaging inspection (see 4.7)

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105, using the following.

Inspection or test	Level	AQL
Product examination	II	1.0
Leakage test	II	1.0
Packaging inspection	S-4	4.0

4.5 Product Examination. Visually, dimensionally, and manually examine each torch handle to determine conformance with the requirements of 3.2 through 3.2.7, and 3.4 through 3.6. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from identified defects. Dimensional examination includes measuring dimensions including weights, as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the torch handle and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.2.3), shall be cause for rejection.

4.6 Tests. Tests shall to be conducted using pressure gauges calibrated to within three percent of full scale at all readings in the upper and lower 1/4 portions of the scale and shall be accurate to within two percent of full scale value at all readings within the center 1/2 portion of the scale. Flow meters shall be accurate within three percent of full scale reading. For flow rate tests the pressure gauges shall be located between the handle inlets and the flow meters. Failure of the handle to meet all the requirements herein shall be cause for rejection.

4.6.1 Oxygen flow rate. Test may be conducted using oil-free air or nitrogen instead of oxygen. Gradually increase flow through the oxygen control valve. Flow shall reach 450 scfh with the inlet pressure at or less than 90 psig (see 3.3.1).

4.6.2 Acetylene flow rate. Test may be conducted using oil-free air or nitrogen. Gradually increase flow through the acetylene control valve. Flow shall reach 25 scfh with the inlet pressure at or less than 10 psig (see 3.3.2).

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4.6.3 Leakage. Attach the specified brand name welding head to the handle. Attach a supply hose to the oxygen control valve inlet and close the valve by tightening to no more than 20 inch-pounds of torque. Submerge the handle in clean water. Apply a minimum of 50 psig pressure to the control valve. There shall be no formation of bubbles for a continuous ten second period. Open the control valve. There shall be no leakage from the entire assembly in a continuous ten second period. Remove the supply hose from the oxygen inlet and attach the hose to the acetylene control valve inlet. Repeat the leakage test. There shall be no leakage during each of two continuous ten second leak tests (see 3.3.3 and 3.4.2).

4.7 Packaging inspection. The torch handle shall be inspected before and after packaging to determine compliance with Section 5 herein.

5 PACKAGING

5.1 Preservation and packing. Unless otherwise specified (see 6.2.1) preservation, level A, and packing, levels A or B shall be in accordance with the requirements of MIL-W-45562. Unit pack quantity shall be 1 each.

5.2 Marking. Unless otherwise specified (see 6.2.1) marking of unit intermediate and shipping containers shall be in accordance with MIL-STD-129.

5.3 Industrial. When specified (see 6.2.1), preservation, packing and marking shall be in accordance with ASTM D3951-82. Unit pack quantity shall be 1 each.

5.4 Thread protection. Threads on the handle, both gas inlets and the attachment end shall be protected from damage during handling, storage and shipments with plastic caps conforming to MIL-C-5501.

6 NOTES

6.1 Intended use. Torch handles procured in accordance with this specification are used as replacements for original equipment torches used by the field military.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Brand name and model number or part number of the handle that is being replaced (see 3.2).

c. Brand name and model number of the welding head or the cutting attachment that the original equipment handle is used with. Either one will suffice since the interfacing connection are identical (see para 3.2).

d. Product identification, if different (see 3.5).

e. First article inspection, if different (see 4.3).

f. Level of preservation, packing and marking (see 5.1, 5.2, and 5.3).

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6.3 First article. When a first article inspection is required, the attachments should be a first article sample, a first production item, a sample selected from the first run of production items or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of a specific number of units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering products which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government is presently appropriate for the pending contract.

6.4 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract. (See 3.8)

6.5 Definition.

6.5.1 Standard cubic feet per hour. Standard air is defined in the Compressed Air and Gas Handbook, copyright 1973, by the Compressed Air and Gas Institute, as air at a temperature of 68 degrees Fahrenheit (F), a pressure of 14.70 pounds per square in absolute (psia) and a relative humidity of 36 percent (0.0750 density). Standard cubic feet per hour is the delivery rate of standard air.

6.6 Subject term (key word) listing.

Attachment, cutting
Backfire
Compressed Gas Association
Cutting attachment
Flashback
Oxygen
Welding apparatus

Custodians:

Army - AL
Navy - YD
Air Force - 99

Review activities:

Air Force - 84

Preparing activity:

Army - AL

Project No. 3433-0110

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-45998	HANDLE OXYACETYLENE WELDING AND CUTTING TORCH

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

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